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## 1 FUSELAGE INNOVATIONS' PATENTED DRONES FOR SECURE AGRICULTURE SOLUTIONS

Kochi-based drone startup Fuselage Innovations manufactures agricultural drones and AI solutions for crop growth analysis and predictions...



FIA QD10

**E**stablished in 2020 by Devan and Devika Chandrasekharan, Fuselage Innovations Private Limited focuses on integrating drones and IoT solutions within the agri-tech sector.

Devan explains that the company's name, 'Fuselage Innovations,' is derived from the fuselage—the central structure of an aircraft that houses passengers and cargo. This name symbolises the critical role of coordination among aircraft components, mirroring the company's innovative approach and the integration of various elements in its solutions.

The startup's agricultural drones and AI solutions are designed to assist farmers in planning their activities and following precision farming protocols tailored to specific crops. Beyond agriculture, Fuselage

Innovations also engages in defence projects, collaborating with the Defence Research and Development Organisation (DRDO) and working on initiatives for shipyards.

Under the Make in India initiative, Fuselage Innovations has developed indigenous drones. Their surveillance drone, designed for data analysis, is named Nireeksh AG, while their crop-spraying drone, certified by the Civil Aviation Department, is known as FIA QD10. Additionally, they offer tracking devices to monitor drone performance, with a platform that applies this technology to agricultural machinery. This system prevents unauthorised transfers between states and ensures compliance and security. The company has developed an integrated approach between its two drones, patenting

the entire product design and device.

Their Nireeksh drone has a multi-spectral range of sensors that capture plant data and processes it into 2D and 3D maps. These maps are further analysed to create index maps, such as vegetation index data, allowing visualisation and understanding of crop growth stages and stress levels. Based on this data, they design and share prescriptions with the spraying drone FIA QD10, which accurately controls and targets applications to address specific crop deficiencies. While the FIA QD10 can operate independently with high accuracy, it functions effectively in integrated operations.

Fuselage Innovations develops the frame structures, power distribution boards, CAN boards, propellers, tanks, envelopes, and the software platform in-house. However, some critical components that must meet aeronautical standards rely on imports. They handle approximately 60% of their development in-house, outsource around 30%, and import the remaining 10%.

The company's primary target audience includes stakeholders from the food and cash crop sectors, particularly those involved in mono-crop-based farming. Additionally, they focus on stakeholders in the food and cash flow industry who align with farmers' producer organisations (FPO)s. "We also collaborate